

1. Positive Indices

- A** 1 a^4 2 m^3 3 a^2b^3 4 k^3m^2 5 3^2 6 5^4 7 $4y^2$
 8 64 9 16 10 75
- B** 1 a^5 2 m^6 3 y^3 4 k^7 5 a^{14} 6 y^6 7 a^4
 8 m^5 9 k^3 10 y^6 11 m^4 12 x 13 a^{12} 14 y^6
 15 m^{10} 16 x^{24} 17 k^{20} 18 a^{36} 19 m^{11} 20 a^8
 21 a^8 22 x^3 23 $21m^6$ 24 $12y^7$ 25 $9a^2$
 26 $125x^3$ 27 $2m$ 28 $5y^2$ 29 $6x^4$ 30 $12k^{12}$
 31 $18m^7$ 32 a^4b^2 33 a^3y^5 34 $5a^2b^3$ 35 $10m^3n^7$
 36 $21a^5k^6$ 37 $4x^6$ 38 ab^2 39 x^4y^4 40 $4y^3$
 41 a^6b^3 42 $x^{12}y^6$ 43 $4a^6m^4$ 44 m^3 45 $3a^2$
 46 m^{14} 47 a^8 48 $54k^{10}$ 49 $4x^2$ 50 a^5
- C** Riddle: It has just become engaged.
 1 a^5b^{11} 2 $10a^2b^8$ 3 $6a^3b^5$ 4 y^2 5 $2a^4m^8$
 6 $6x^2y^2$ 7 $9a^4b^6$ 8 $16x^4y^6$ 9 $72a^5$ 10 $27m^3$
 11 $3x^4$ 12 a^8 13 $8a^{11}$ 14 m^{32} 15 $6a$ 16 $6a^2b^{10}$
 17 $12a^6$ 18 $30a^7b^3$ 19 $40a^5b^7$ 20 $a^{20}b^5$ 21 $50y^2$
 22 $4a^2$ 23 $3m^{10}$ 24 $2a^7b^9$

2. Negative Indices

- A** 1 a^8 2 m^{12} 3 $6y^{11}$ 4 $16t^8$ 5 y^4 6 m^6 7 $5x^4$
 8 $5a^4$ 9 x^{12} 10 $25a^8$
- B** 1 $\frac{1}{m^2}$ 2 $\frac{1}{a^4}$ 3 $\frac{1}{x}$ 4 $\frac{2}{y^2}$ 5 $\frac{5}{m^3}$ 6 $\frac{y}{x^2}$ 7 x^{-3}
 8 m^{-6} 9 a^{-4} 10 $4a^{-3}$ 11 $5x^{-2}$ 12 $2xy^{-4}$
 13 m^{-6} 14 a 15 x^{-2} 16 k^5 17 m^{-2} 18 y^{-8}
 19 a^{-6} 20 k^{12} 21 $15a^{-2}$ 22 28 23 $8x^{-3}$
 24 $3y^{-1}$ 25 m^2 26 $\frac{1}{k^2}$ 27 $\frac{1}{a^5}$ 28 $\frac{1}{m^2}$ 29 $\frac{1}{y}$
 30 $\frac{15}{a^2}$ 31 $\frac{4}{m^3}$ 32 $\frac{3}{x}$ 33 $\frac{1}{m^3}$ 34 $\frac{4}{a^2}$ 35 $\frac{1}{3x^2}$
 36 $\frac{1}{a^2}$ 37 $\frac{1}{ab^2}$ 38 $\frac{y}{x^3}$ 39 $\frac{1}{a^2}$ 40 $\frac{1}{a^2}$
- C** 1 $\frac{2}{x}$ 2 $\frac{x}{2}$ 3 $9y^2$ 4 1 5 $\frac{1}{a}$ 6 $\frac{1}{m^2}$ 7 $\frac{81}{y^8}$
 8 $\frac{1}{x^6}$ 9 $\frac{b^3}{8a^3}$ 10 $4ab^3$ 11 $32m^3$ 12 3 13 $4a^2$
 14 a^9 15 m^3 16 $\frac{a}{b^3}$ 17 $\frac{y^5}{x^2}$ 18 ab 19 $\frac{y^2}{x^5}$
 20 $\frac{3a^7}{b^7}$

3. Fractional Indices

- A** 1 m^9 2 $6a^8$ 3 y^5 4 $2x^4$ 5 x^3y^3 6 a^8 7 $27y^8$
 8 m^{-4} 9 y^2 10 $6x^{-2}$
- B** 1 5 2 7 3 2 4 5 5 4 6 3 7 8 8 2 9 $x^{\frac{1}{2}}$
 10 $m^{\frac{1}{3}}$ 11 $36^{\frac{1}{2}}$ 12 $a^{\frac{1}{4}}$ 13 $64^{\frac{1}{3}}$ 14 $x^{\frac{1}{5}}$ 15 $a^{-\frac{1}{2}}$
 16 $m^{-\frac{1}{3}}$ 17 x^2 18 y^2 19 y 20 m^4 21 m
 22 a^2 23 y 24 ab^2 25 xy^4 26 $3x^2$ 27 $2x^3$
 28 $2m$ 29 $2a^2$ 30 $4m^5$ 31 $2a^3$ 32 $15a$ 33 $18k$
 34 $2ab^3$ 35 $2xy$ 36 $2y^5$
- C** 1 27 2 8 3 9 4 32 5 $\frac{1}{3}$ 6 $\frac{1}{3}$ 7 $m^{\frac{1}{3}}$ 8 $m^{\frac{1}{3}}$
 9 $a^{\frac{1}{3}}$ 10 $a^{\frac{1}{3}}$ 11 $x^{\frac{1}{3}}$ 12 $x^{\frac{1}{3}}$ 13 m^2 14 $4a^2$
 15 $2x^4$ 16 y^2 17 $2y^4$ 18 24 19 $64a^6$ 20 $32x^5$
 21 $4y^6$ 22 $125m^9$ 23 $8x^6$ 24 $\frac{1}{3y}$ 25 $\frac{1}{3a^2}$
 26 a^3b^4 27 $a^{\frac{1}{2}}$ 28 $\frac{1}{2}$ 29 $9a^{\frac{1}{2}}$ 30 $a^{\frac{1}{2}}$

4. Indices—Review

- A** 1 a^{12} 2 m^9 3 x^4 4 y^2 5 k^{12} 6 m^{14} 7 $9x^2$
 8 $16a^4$ 9 y^8 10 m^2

- B** 1 $2a^7$ 2 $8m^7$ 3 $35k^{10}$ 4 $32y^9$ 5 $6m^2$ 6 $5a^4$
 7 $4x^5$ 8 $4y^8$ 9 $8x^9$ 10 $16m^{10}$ 11 $27y^6$ 12 a^4b^2
 13 $x^{12}y^6$ 14 m^2n^8 15 a^5b^2 16 x^2y^5 17 m^6n^8
 18 ay^7 19 x^5 20 k^2m^2 21 $4m^6$ 22 a^4b^3
 23 $4a^4b^2$ 24 $125x^9y^5$ 25 m^{14} 26 a^{15} 27 $30a^5b^5$
 28 $3xy$ 29 $81m^4$ 30 y^{14} 31 $6a^4b^2$ 32 $30a^2b^2$
 33 a^2 34 $2k^3$ 35 $45y^{10}$ 36 $16m^6n^4$ 37 1
 38 $15a^3b^7$ 39 a^3m 40 $3m^5$ 41 $2m^3$ 42 $4x$
 43 y^{-2} 44 2 45 a^6 46 k^{-1} 47 $24m$ 48 $3x^{-4}$
 49 $5x^{-5}$ 50 $x^{-2}y^3$ 51 m^8 52 x 53 x^{-2} 54 $2x^3$
 55 x^2y^3 56 y^4 57 a 58 m^4

Riddle: Because he had nobody to go with.

- C** 1 a^2 2 $6y^{10}$ 3 $80y^{11}$ 4 x 5 $2a^8$ 6 $3m$ 7 $45m$
 8 $2m^3$ 9 5 10 $\frac{4a}{3b}$ 11 $\frac{y}{3x^2}$ 12 a^3 13 $3x^2y^3$
 14 $2xy^2$ 15 $12y$ 16 $16a^8$ 17 $a^{-5}b^{-2}$ 18 xy^{-2}
 19 $6m^6$ 20 $5x^2y^2$

5. Simplifying Surds

- A** 1 $2\sqrt{2}$ 2 $2\sqrt{3}$ 3 $5\sqrt{2}$ 4 $3\sqrt{2}$ 5 $4\sqrt{3}$ 6 $3\sqrt{3}$
 7 $2\sqrt{5}$ 8 $5\sqrt{3}$ 9 $2\sqrt{6}$ 10 $7\sqrt{2}$ 11 $3\sqrt{5}$
 12 $2\sqrt{11}$ 13 $3\sqrt{6}$ 14 $6\sqrt{2}$ 15 $3\sqrt{7}$ 16 $10\sqrt{5}$
 17 $3\sqrt{10}$ 18 $9\sqrt{2}$ 19 $3\sqrt{11}$ 20 $7\sqrt{7}$
- B** 1 18 2 12 3 32 4 50 5 20 6 48 7 80
 8 75 9 28 10 99 11 $\sqrt{2}$ 12 $\sqrt{13}$ 13 $\sqrt{34}$
 14 $2\sqrt{3}$ 15 $2\sqrt{6}$ 16 $3\sqrt{3}$ 17 2 and 3
 18 5 and 6 19 4 and 5 20 9 and 10 21 6 and 7
 22 12 and 13 23 3 24 7 25 5 26 12
 27 3 28 2
- C** 1 $2\sqrt{2}$ 2 $2\sqrt{5}$ 3 $3\sqrt{2}$ 4 $5\sqrt{2}$ 5 F 6 F 7 T
 8 T 9 $8\sqrt{2}$ 10 $10\sqrt{3}$ 11 $18\sqrt{2}$ 12 $18\sqrt{5}$
 13 $a\sqrt{b}$ 14 $y\sqrt{x}$ 15 $2a\sqrt{a}$ 16 $3x\sqrt{y}$
 17 $\sqrt{3}, 2, \sqrt{5}, \sqrt{7}, 3, 4$ 18 $4, 3\sqrt{2}, \sqrt{20}, 2\sqrt{6}, 5$

6. Addition and Subtraction of Surds

- A** 1 $2\sqrt{2}$ 2 $2\sqrt{3}$ 3 $3\sqrt{2}$ 4 $5\sqrt{3}$ 5 $3\sqrt{2}$ 6 $5\sqrt{5}$
 7 $\sqrt{2}$ 8 $3\sqrt{5} + \sqrt{2}$ 9 $2\sqrt{2} - \sqrt{3}$ 10 $\sqrt{5} - \sqrt{6}$
- B** 1 $3\sqrt{2}$ 2 $5\sqrt{2}$ 3 $3\sqrt{3}$ 4 $\sqrt{3}$ 5 $4\sqrt{3}$ 6 $4\sqrt{3}$
 7 $5\sqrt{3}$ 8 $5\sqrt{3}$ 9 $3\sqrt{5}$ 10 $5\sqrt{2}$ 11 $6\sqrt{3} + 2$
 12 $8\sqrt{3} + 5\sqrt{2}$ 13 $5\sqrt{6} - 5\sqrt{7}$ 14 $2\sqrt{3} - 3\sqrt{2}$
 15 $\sqrt{5}$ 16 $2 + \sqrt{2}$ 17 $\sqrt{3}$ 18 $5\sqrt{10}$ 19 $2\sqrt{3}$
 20 $3\sqrt{2}$ 21 $\sqrt{3} + 2$ 22 $5\sqrt{11}$ 23 $2 + \sqrt{3}$
 24 $2\sqrt{5}$ 25 $\sqrt{3}$ 26 0 27 $6\sqrt{3}$ 28 $20\sqrt{2}$
- C** 1 $2\sqrt{6}$ 2 $12\sqrt{2}$ 3 $\sqrt{2}$ 4 $12\sqrt{3}$ 5 $7\sqrt{2}$ 6 $\sqrt{7}$
 7 $7\sqrt{5}$ 8 $10\sqrt{2} + 10\sqrt{3}$ 9 $\sqrt{2}$ 10 $\sqrt{6}$ 11 $11\sqrt{2}$
 12 $8\sqrt{2}$ 13 $8\sqrt{3}$ 14 $\sqrt{10}$ 15 $6\sqrt{3}$ 16 $5\sqrt{x}$
 17 $2\sqrt{x} - \sqrt{y}$ 18 $\sqrt{a} + a$ 19 $5\sqrt{b} - 7$ 20 $\sqrt{c}$
 21 T 22 F 23 F 24 F

7. Multiplying Surds

- A** 1 2 2 3 3 5 4 $\sqrt{6}$ 5 $5\sqrt{5}$ 6 $\sqrt{35}$ 7 $\sqrt{30}$
 8 $\sqrt{30}$ 9 $\sqrt{110}$ 10 6 11 4 12 6 13 12
 14 12 15 12 16 $10\sqrt{15}$ 17 $20\sqrt{6}$ 18 -2
 19 -6 20 $-2\sqrt{2}$
- B** 1 $2\sqrt{3}$ 2 $3\sqrt{2}$ 3 $60\sqrt{2}$ 4 78 5 4 6 $3\sqrt{3}$
 7 8 8 18 9 12 10 50 11 20 12 24
 13 $135\sqrt{5}$ 14 24 15 $36\sqrt{2}$ 16 $16\sqrt{2}$ 17 $6\sqrt{2}$
 18 10 19 42 20 $30\sqrt{3}$ 21 12 22 $4x$ 23 $9y$